

Anatomy

(Scheduled Event - 30 Minutes)



Description: Participants will be assessed on their understanding of the anatomy and physiology of the human digestive system.

Participants per assigned Team Number: 2

- If your school has 1 team you will send 2 students
- If your school has 2 teams you will send 2 students per team number; team numbers may not intermix
- If your school has 3 teams you will send 2 students per team number; team numbers may not intermix

Teams must bring: Pen/Pencil

Teams may not bring:

- Notes or guides of any kind.

Optional - Teams may bring: Four function calculator

Safety Requirements: None

The Competition: This event may be administered as a written test or as a series of stations which can include diagrams, models, specimens, data analysis, experiments, and problems for students to solve. Content topics will include but are not limited to:

1. Organization and Anatomy of the Digestive System

- Functions of the digestive system.
- Distinguish between digestive tract organs and accessory digestive organs.
- Identify the major digestive organs and describe their structure and function.

2. Digestive Processes

- Trace the pathway of food through the digestive system.
- Compare and contrast mechanical and chemical digestion.
- Describe the roles of saliva, stomach acid, bile, and digestive enzymes.
- Explain how carbohydrates, proteins, and fats are broken down into absorbable molecules.

3. Absorption, Transport, and Regulation

- Describe how nutrients and water are absorbed and transported throughout the body.
- Compare the roles of the small intestine and large intestine in absorption.
- Explain how digestive structures support absorption.
- Describe how the nervous system and digestive organs regulate and coordinate digestion, including hunger and satiety signals.

4. Nutrition and Digestive Health

- Identify the major nutrient groups: carbohydrates, proteins, fats, vitamins, minerals, and water.
- Describe the primary functions of each nutrient group.
- Explain the characteristics of a balanced diet and its relationship to digestive health.

5. Digestive System Disorders

- Describe the causes, symptoms, prevention, and basic treatment of acid reflux (GERD), lactose intolerance, constipation, diarrhea, peptic ulcers, and appendicitis.

6. Data Analysis and Applications

- Interpret diagrams, charts, tables, models, and experimental data related to digestion and digestive health.
- Apply knowledge of digestive anatomy and physiology to solve problems and answer scenario-based questions.

Examples:

- Identify major digestive organs from diagrams or models
- Distinguish between **digestive tract organs** and **accessory organs** using diagrams or tables.
- Trace the pathway of food through the digestive system from ingestion to elimination.
- Interpret diagrams showing how food moves through the digestive tract.
- Identify where mechanical digestion and chemical digestion occur.
- Compare the roles of - chewing, stomach churning, enzymes, acids, bile
- Analyze models showing how carbohydrates, proteins, and fats are broken down into smaller molecules.
- Match digestive enzymes and substances to their functions:
- Interpret charts or tables showing nutrient breakdown and absorption.
- Compare the functions of the small intestine and large intestine using diagrams or data.
- Analyze diagrams of intestinal structures (e.g., villi) to explain how they increase nutrient absorption.
- Interpret experimental data related to digestion, such as:
 - effect of temperature on enzyme activity
 - effect of pH on digestion
 - rate of food breakdown
 - Analyze nutrition labels or food tables to identify: carbohydrates, proteins, fats, vitamins, minerals, water
 - Compare diets and determine which provides better nutritional balance.
 - Interpret charts showing relationships between diet, hydration, and digestive health.
- Analyze case studies involving digestive disorders, including: acid reflux (GERD), lactose intolerance, constipation, diarrhea, peptic ulcers, appendicitis
 - Use symptoms and scenario information to identify possible digestive problems.
 - Recommend basic prevention strategies related to digestive health, such as: fiber intake. Hydration, balanced nutrition, healthy eating habits
- Apply digestive system knowledge to real-world scenarios (e.g., "Why does a person with lactose intolerance experience symptoms after consuming milk?").

Scoring:

- Points will be awarded for each correct response.

Tiebreakers:

- Ties will be broken by the accuracy or quality of answers to select questions chosen by the event leader prior to the competition.

Resources:

- The Human Body Book by Steve Parker
- [Biology for Kids](#)
- Science Teachers & Librarians
- Get permission from your parents/coach to use AI to create a study guide using the rules provided here

Allowed Calculator

This is the acceptable calculators for all Division A events.

Class I - Stand-alone non-graphing, non-programmable, non-scientific 4-function or 5-function calculators. These are the most basic types of calculators. They are limited to the four basic mathematical functions and square roots. They are often cheap, costing as low as a dollar online.



Safety Glasses

Indirect Vent Goggles or Safety glasses are a mandatory safety requirement for some events. It is important to choose the type of protection specific to the task. Competitors are responsible for providing their own protective eyewear. Teams will not be allowed to compete without adequate eye protection.

Category C: Indirect Vent Goggles



Category B: Safety glasses



Compliant Eyewear Categories:

CATEGORY B

- Description: Impact protection. They provide protection from a high inertia particle hazard (high mass or velocity)
- Corresponding ANSI designation/required marking: Z87+.
- Category B eyewear can look either like goggles or safety glasses, but must have the Z87+ marking embossed on them.

CATEGORY C

- Description: Indirect vent chemical/splash protection goggles. These seal tightly to the face completely around the eyes and have indirect vents constructed so that liquids do not have a direct path into the eye (or no vents at all). If you can see through the vent holes from one side to the other, they are NOT indirect vents
- Corresponding ANSI designation/required marking: Z87 (followed by D3 is the most modern designation but, it is not a requirement)

Examples of Non-Compliant Eyewear:

- Face shields/visors are secondary protective devices and are not approved.
- Prescription Glasses containing safety glass should not be confused with safety spectacles. "Safety glass" indicates the glass is made to minimize shattering when it breaks. Unless these glasses bear the Z87+ mark they are not approved for use.

Notes:

1. A goggle that bears the Z87+ mark and is an indirect vent chemical/splash protection goggle will qualify for both Category B & C
2. VisorGogs do not seal completely to the face, but are acceptable as Type C goggles